7. Battery Subsystems

7.1 Design Process

7.1.1 Battery Relevant Rules

A large portion of the 2023 FSAE rules [2] pertain to accumulator design, specifications, and the implementation of the Tractive/HV System of the vehicle. As such, the rules pertaining to the Battery sub-team were first logged and understood. The ruleset defines rules regulating both the documentation and competition requirements (including technical inspection testing) and specific rules pertaining to component design.

The ruleset was combed over by the entire sub-team, involving paraphrasing all pertinent rules and identifying potential overlaps with other sub-teams. To ensure adherence to the rules throughout the design phases and iterations a practice rules check was performed on November 1st. During the rules check, both completion and rules compliance were evaluated for each pertinent rule. The following excel table includes a portion of the 60 identified rules, including the summarized rule, sub-team overlap, and result of the 11/1 rules check. The full BRR spreadsheet is in Appendix _.

* indicates includes all subordinates	Battery Relevant Rules				
	Rule #	**Relevant Scope Items**	**Content**	**Subteam Overlap**	**Rules Check 11/1**
Documentation, Competition, Tech Inspection	F.10.1.*	Documentation Event, Tech	SES, Accumulator Tech	C	F.10.1.1a Fail, no FEA
	IN.4.*	Tech	Accumulator Tech, EV Tech what to bring & procedures	E	
	IN.8.1	Tech	Bring accumulator samples	C,S,A	
	IN.9.*	Tech	Coolants can't leak	C,S	
	IN.11.*	Tech	Rain test	E	
	IN.14.*	Tech	Post-tech allowed mods	All	
	S.2.*	Documentation Event	Documentation Event	All	
	S.3.*	Cost Event	Cost Event	All	
	S.4.*	Design Event, Competition	Design Event	All	
	D.2.*	Competition	Paddock rules	All	
	D.13.*	Efficiency Event, Competition	Efficiency Event		
	EV.2.*	Documentation Event	Electrical System Form (ESF); Format	E	
	EV.10.*	Competition	Fixing/moving accumulator, charging, activation	All	
Design	EV.1.*	All Electric	TS/GLV Definition, Accumulator Definition	E,P	pass
	EV.2.*	ESF	ESF	E,P	pass
	EV.3.2.*	Manufacturing	PPE, Insulated Tools		pass
	EV.4.1.*	Electrical Configuration	Power/Voltage/Regen/Reverse LImits	P	pass
	EV.4.(2/3/4).*	Energy Meter	Energy Meter packaging		need to design energy meter enclosure
	EV.5.3.*	Accumulator Container	Removability, Holes, Ventalation		EV.5.3.7 need label
	EV.5.8	Enclosures	Labels, Conductivity to GLV	P	pass
	EV.5.10.*	Charging Cart	Use, Design, Brake		EV.5.10.4c brake strength may be insufficient
	EV.6.1.*	Segments	Energy limits, etc	P	EV.6.1.2 waiting on clarification for segment energy
	EV.6.2.*	Electrical Configuration	Ratings, Insulation, Solder		EV.6.2.1 have not selected all components yet EV.6.2.5 just need to double check with electronics
	EV.6.3.*	Maintenance Plugs	Requirements		FAIL ALL

Figure 50: Battery Relevant Rules Spreadsheet

7.1.2 Battery Sub-Team Scoping

Prior to beginning their design, the Battery sub-team defined the scope of their design project after thoroughly reviewing the rules. The scope is divided into four major sections: accumulator and container, segments, and cells, BMS and charging, and control and HVILs. The following table defines the scope as observed by the four previous partitions, with highlighted entities designating time-intensive tasks.

Table 10: Battery Sub-Team Scoping

Scope			
Accumulator Container	**Segments & Cells**	**BMS & Charging**	**Control & HVILs**
Structure & Material	Cell selection	BMS Selection/design	AIR selection
Cooling (maybe aero too)	Cell Configuration	BMS Configuration	HVIL electronics
Mounting & Removal	Segment Structure	Charger selection/design	
Waterproofing	Maintainence Plugs	BMS Tuning	Precharge
Insulation	Tab Connection	IMD	Discharge
	Internal TS Conductors	Internal GLV Conductors	*Energy Meter*
	Fuses	Temperature Sensors	
	Connectors	Charging Cart	

7.1.3 Iterative Design Process

An iterative design process was leveraged like other Mechanical sub-teams (Suspension and Aerodynamics) to effectively design the accumulator and ensure proper integration within the Tractive System. *Figure 51* details the order and iteration of the design process.

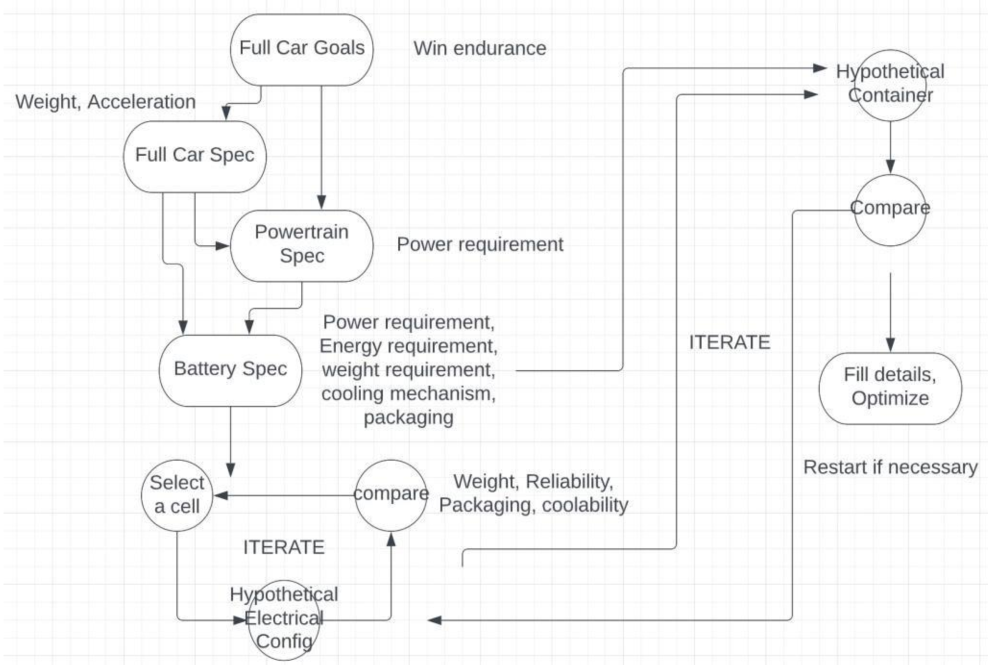


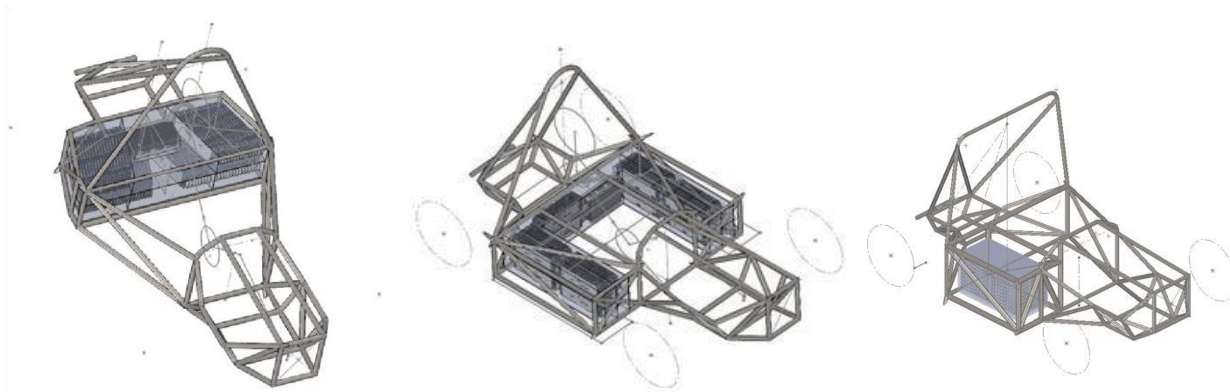
Figure 51: Iterative Design Process

7.2 Accumulator Shape and Location Determination

To effectively determine optimal positioning of the accumulator, several options were considered. The accumulator placement is highly critical, as due to the large size and mass attributed with the accumulator, it will be one of the driving factors in the overall CG balance of the full vehicle. The following three configurations listed in **Table 11** were selected to further analyze. CAD models of the accumulator and the resulting position of the full vehicle are also shown below.

Table 11*:* Accumulator Shape and Possible Locations

Accumulator Shape and Location Placements	
CONFIGURATION	**VEHICLE LOCATION**
Sidewinder	Beside Driver
U-Pack	Around Driver
Y-Long	Behind Driver (in front of rear suspension box)



Figures 52-54: Possible Accumulator Shapes and Locations within Corresponding Chassis

The above configurations were then evaluated through comparison using a decision matrix. Both pouch and cylindrical cell configurations were considered for each of the three accumulator shapes at this stage in the design process. The matrix compared CG deviation of each pairing, the complexity involved in implementing within the full vehicle, the complexity of the battery, the effect on full vehicle weight, and the resulting experience for the driver. Implementation complexity, CG deviation, and full vehicle weight effect were given the largest weightings to prioritize a smooth implementation and ensure minimization of endurance lap times by means of proper CG placement and weight minimization.

Table 12*: Accumulator Shape and Location Decision Matrix*

Battery Location Final Matrix

Percentage	25%	26%	19%	25%	5%			
Criteria	**CG Deviation**	**External Complexity**	**Battery Pack Complexity**	**Full Car Mass**	**Driver Experience**	**Score**	**Normalized Score**	**Spread**
Pouch (Y-Long) Behind Driver - Lateral	10.00	7.50	6.80	7.94	7.00	44.00	10.00	0.00%
Pouch (U-Pack) U-Shape Behind Driver	7.54	4.95	2.00	7.06	5.00	32.93	7.48	100.00%
Pouch (Sidewinder) Beside Driver	8.56	4.30	9.40	8.02	2.00	38.19	8.68	52.48%
Cylindrical (Y-Long) Behind Driver - Lateral	6.82	7.05	6.60	9.87	7.00	43.81	9.96	1.72%
Cylindrical (U-Pack) U-Shape Behind Driver	8.07	4.95	2.00	8.75	5.00	36.80	8.36	65.04%
Cylindrical (Sidewinder) Beside Driver	7.11	4.60	9.60	10.00	2.00	43.31	9.84	6.23%

Through the decision matrix, the Y-long accumulator shape was determined to best suit the design considerations of the full vehicle. The Sidewinder configuration also scored highly; however, it was not chosen due to low scoring in implementation complexity and CG deviation. Between the two Y-long options (cylindrical cells and pouch), the cylindrical cell design was chosen to proceed despite having the lower normalized score due to higher individual scoring in full car weight result despite worse CG deviation.

7.3 Cell, Configuration Selections

7.3.1 TESLA Battery Sponsorship

In adherence with their company goal to "accelerate the world's transition to sustainable energy", TESLA [14] decided to offer up to 10 [kWh] of battery cells and up to 30 ASIC chips from ADI [15] or a custom BMS developed by TI [16]. To be eligible, each team applying was required to complete a design proposal addressing the needs of the team and the benefits of receiving the TESLA sponsorship. Eligible teams were also required to provide registration documentation for Formula SAE Michigan (June 2022) or Formula Student Germany (August 2022), of which Texas A&M has already registered for Formula SAE Michigan.

The Texas A&M FSAE Electric team was awarded the TESLA sponsorship. In addition to the cell variety offered directly from TESLA, a $5,000 discount was also offered for hardware ordered from Enepaq [17]. Thus, all viable options for cells were compared to ensure all valid configurations were thoroughly evaluated.

Table 13: TESLA Battery Sponsorship Cell Options

TESLA Cell Options
21700 Cylindrical [x]
18650 Cylindrical [x]
13 Ah Pouch [x]
20 Ah Pouch [x]

$5,000 Enepaq Hardware Order Discount

7.3.2 Cell Selections Utilizing Python

To compare cell configurations, a Python script was generated. The script reads in information from the cell manufacturer's data sheets and considers 54 total cells. From the cell data, the mass, energy, maximum power, nominal power, voltage range, and volume were all subsequently calculated. After calculation, the script ranks the top 8 valid configurations by least to greatest mass. The following setpoints were also provided to the script to ensure adherence to the energy requirements received from the Powertrain sub-team.

Table 14: Full-Pack Configuration Setpoints Used in Python Cell Comparison Script

Full-Pack Configuration Setpoints	
PARAMETER	**INPUT RANGE**
Voltage Range [V]	450-300 // 300-150
Peak Power Minimum [kW]	80
Energy Range [kWh]	7-8

```
1) IMR (10) [100]s<7>p: MASS= 32.06[kg], ENERGY= 7.77[kWh], MAX POWER= 102.9[kW]
NOMINAL POWER= 51.8[kW], VOLTAGE RANGE= 250.0 <V< 420.0, VOLUME= 0.14[m^3]

2) P28A (7) [100]s<7>p: MASS= 32.2[kg], ENERGY= 7.06[kWh], MAX POWER= 102.9[kW]
NOMINAL POWER= 50.4[kW], VOLTAGE RANGE= 250.0 <V< 420.0, VOLUME= 0.14[m^3]

3) VTC6 (2) [100]s<7>p: MASS= 33.25[kg], ENERGY= 7.86[kWh], MAX POWER= 161.7[kW]
NOMINAL POWER= 39.312[kW], VOLTAGE RANGE= 250.0 <V< 420.0, VOLUME= 0.14[m^3]

4) Tesla Sponsorship P42A (8) [100]s<5>p: MASS= 33.9[kg], ENERGY= 7.56[kWh], MAX POWER= 94.5[kW]
NOMINAL POWER= 54.0[kW], VOLTAGE RANGE= 250.0 <V< 420.0, VOLUME= 0.1[m^3]

5) 40T (46) [100]s<5>p: MASS= 34.099999999999994[kg], ENERGY= 7.2[kWh], MAX POWER= 105.0[kW]
NOMINAL POWER= 90.0[kW], VOLTAGE RANGE= 250.0 <V< 420.0, VOLUME= 0.1[m^3]

6) 25R (41) [100]s<8>p: MASS= 35.04[kg], ENERGY= 7.37[kWh], MAX POWER= 336.0[kW]
NOMINAL POWER= 58.9824[kW], VOLTAGE RANGE= 250.0 <V< 420.0, VOLUME= 0.16[m^3]

7) 25R (4) [100]s<8>p: MASS= 36.0[kg], ENERGY= 7.2[kWh], MAX POWER= 336.0[kW]
NOMINAL POWER= 57.6[kW], VOLTAGE RANGE= 250.0 <V< 420.0, VOLUME= 0.16[m^3]

8) Tesla Sponsorship COSMX 20Ah (53) [95]s<1>p: MASS= 44.65[kg], ENERGY= 7.03[kWh], MAX POWER= 359.1[kW]
NOMINAL POWER= 140.6[kW], VOLTAGE RANGE= 285.0 <V< 399.0, VOLUME= 0.02[m^3]
```

Figure 55: Python Script Cell Evaluation Results

From the results of the script (*Figure 55*), the selected finalists were the Molicel P42A 21700 (cylindrical) [17] and the A123 31Ah (pouch) [18] which was utilized in AME22. These cells were chosen due to them excelling in total weight, whilst still maintaining the 80 [kW] power requirement from Powertrain. Due to the team being awarded the TESLA sponsorship, the decision was made to proceed with using the TESLA supplied 21700 P42A Cylindrical cells. The P42A cells are suited for our FSAE application due to high current capabilities, low heat generation, high energy density, and excellent packageability due to their circular cross section. The following table lists the technical specifications of the Molicel P42A cells.

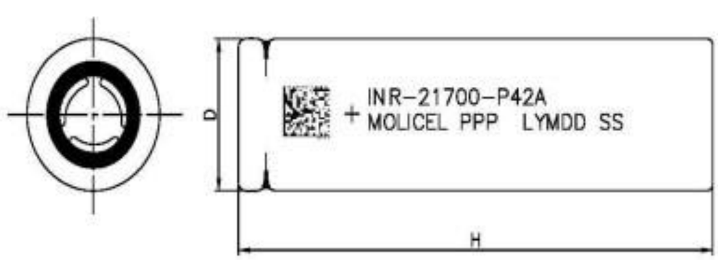


Figure 56: Molicel INR-21700-P42A

Table 15: Molicel INR-21700-P42A Technical Electrical Specifications

■ **CELL CHARACTERISTICS**

Capacity	Typical	4200 mAh 15.5 Wh
	Minimum	4000 mAh 14.7 Wh
Cell Voltage	Nominal Charge Discharge	3.6 V 4.2 V 2.5 V
Charge Current	Standard	4.2 A
Charge Time	Standard	1.5 hr
Discharge Current	Continuous	45 A
Typical Impedance	AC (1 KHz) DC (10A/1s)	10 mΩ 16 mΩ
Temperature	Charge Discharge	0°C to 45°C -40°C to 60°C
Energy Density	Volumetric Gravimetric	615 Wh/l 230 Wh/kg

7.3.3 Configuration Selection

Having chosen the Molicel P42A Cells to use within the accumulator, the sub-team then worked on determining a suitable configuration of the cells. To reach the target voltage of approximately 400 [V], the cells will be used in groupings of parallel series branches of cells. Increasing the number of cells in series increases the maximum achievable voltage. Increasing the number of parallel branches of series cells increases the total energy of the accumulator pack, allowing higher levels of power to be sustained for greater time periods. Using the pack principles, the following configurations were evaluated.

Table 16: Series/Parallel Configurations Evaluated for *Molicel P42A*

		Wh Capacity	Wh Capacity	Wh Capacity	Cell mass	
		15.13	15.5	14.7	0.07	
Pack Config	**Cells**	**Energy Derated**	**Energy Typical**	**Energy Minimum**	**Additional Mass**	**Electrical Considerations**
100s5p (10/8/7)	500	7565	7750	7350	0	ideal
104s5p (10/9/7 or 10/8/8)	520	7867.6	8060	7644	1.4	may exceed motor
108s5p (10/9/8)	540	8170.2	8370	7938	2.8	would need to derate

From the configurations above, the 104s5p (5 parallel branches each consisting of 104 cells in series for 520 total cells) was selected as the configuration to move forward with using the Molicel P42A. Despite the 100s5p configuration being considered ideal for the powertrain, the 104s5p was selected to allow for a ceiling over what is required for the powertrain. The 108s5p configuration was not selected due to the pack requiring de-rating to be integrated with the drivetrain components, due to the 400[V] nature of the motor and motor controllers chosen by Powertrain. Thus, because the pack must be de-rated significantly, the advantages of the 108s5p configuration are not carried over into the physical implementation.

The additional four cells in each parallel branch allow for nearly 80 [kW] of power delivery even when the accumulator pack is at its worst-case condition (nearly depleted of energy). Furthermore, the additional weight of the 20 (4 cells across each of the 5 parallel branches) cells only contributes an additional 1.4 [kg] to the total accumulator mass. Due to the minimal weight addition, the sub-team felt as though the addition of the 20 cells was justified, ultimately leading to better lap times in endurance due to enhanced sustainment of pack power.

7.4 Cooling Requirement Modeling

7.4.1 Possible Methods & Python Modeling for Passive Air-Cooling

To model the heat load generated by the cells, Python was used once again to calculate accumulator heat generation. The sub-team set out to validate the effectiveness of solely aircooling the accumulator (passive cooling) as a means of determining whether active air cooling would be required. The following table includes potential solutions addressing cooling.

Table 17: Accumulator (Battery) Cooling Methods and Resulting Heat Paths

Battery Cooling Methods	
Method	**Heat Path**
Indirect Air - Passive	Cell > Interface > Container > Air
Indirect Air - Active	Cell > Interface > Container > Air
Direct Air - Passive	Cell > (heat sink) > Air
Direct Air - Active	Cell > (heat sink) > Air
Liquid	Cell > (Interface) > Water block > Water > Radiator > Air
Thermoelectric	Cell > (interface) > TEC > Heat sink > Air

The benefits of passive air-cooling include avoiding the need for a fan, minimal complexity, and avoiding further power draw from the pack itself to achieve cooling. Using parameters gained from the lap simulation software VIGrade, the following accumulator statistics were generated
using the Python script and a factor of safety of 1.5 for determination of the 'average' quantities.

```
At nominal voltage:
        Pack Average Heat:  1381.89
        Average Power:  19842.95
        Average Current:  61.73
        Internal Resistance per cell:  0.018
        Efficiency(ish):  0.96
```

Figure 57: Python Results for Accumulator Pack Heat Generation [W], Power [W], Current [A], Cell Internal Resistance[Ω], and Efficiency at Nominal Voltage [V]

The Python cooling script was then used to determine the most optimal air stream velocity for air-cooling cells as well as the minimum airflow required utilizing the flat plate convective heat transfer approximation model. The following represents the initial conditions and factor of safety, and the resulting script outputs.

```
degFlow = 45              #degrees of flow attachment around a cell
degFlowFOS = 2            #FOS for degrees of flow attachment, 1 = no FOS
structureOverlap = 5      #mm of segment structure covering cell side
tAmbient = 35             #C, 35C = 95F
tCell = 50                #C, cell steady state temp (set) when heat gen = heat rejection
vFlowSweepHigh = 30       #km/hr  internal flow velocity WILL SWEEP ZERO TO THIS VALUE
vFlowSweepStep = .01      #km/hr  SWEEP INCRIMENT
cellsInRow = 10           #num cells in row in direction of flow
rowsInPack = 50*2         #100 rows because each row is one side of a cell
reynoldsFOS = 5           #unitless, FOS on effectievness of flow for cooling, 1 = no FOS
```

Figure 58: Python Cooling Script Parameters and Factors of Safety used within calculations

```
Best Velocity Sweep:
        Flow Velocity Per Row 1 Sided [km/hr]:  1.85
        Q Heat Transfered per row 1 sided [W]:  13.81
        Heat disparity [W]:  0.01

Minimum Volumetric Flow Rate: [m^3 / s] 0.0101
Mass Flow Rate [kg / s]:  0.0123
```

Figure 59: Passive Air-Cooling Simulated Results using above parameters and factors of safety.

The Python script indicated a necessary airflow of 0.01 [m^3/s] (22 [cfm]). From the lap simulation software VIGrade, it was determined that the minimum volumetric flow rate of the full car was nearly 220 [cfm]. Therefore, using passive air-cooling, the accumulator pack itself has a factory of safety of about 10. Thus, the sub-team concluded that passive air-cooling would be a viable thermal management solution as at an FoS of 10, phenomena such as thermal runaway and current restriction (due to high temperatures) should be mostly avoided.

7.4.2 Aerodynamics Package Requirements for Passive Air-Cooling

To deliver the necessary free-stream airflow to the cells within the accumulator pack, the Aerodynamics sub-team is implementing "scoops" into their aero package which will allow for unobstructed airflow to the accumulator. Due to the Y-Long pack being placed behind the driver and seat, the "scoops" (directed ducts) serve to channel air from the sides of the driver directly to the accumulator at which point the stream will flow through and out the rear of the vehicle. The blow image highlights a portion of the full vehicle aerodynamics package with the "scoops" circled in red.

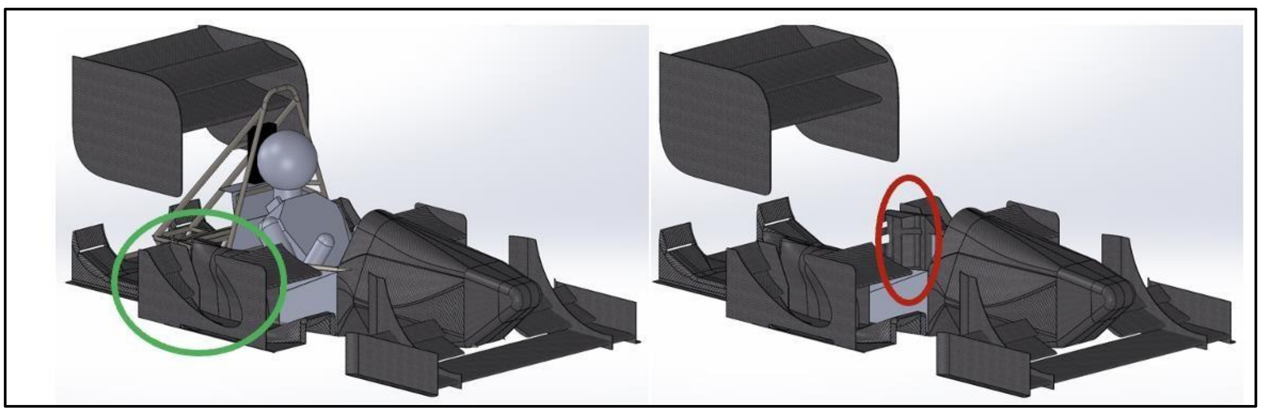


Figure 60 and 61: Full-car vehicle model (with driver and chassis: L, without: R). "Scoop" entrance shown in green and exit to accumulator shown in red.

7.5 BMS Selection and Integration

7.5.1 BMS Requirements and Function

The purpose of the BMS includes providing cell temperature and voltage readings across the accumulator pack, estimating remaining (unused) pack energy in real-time, and regulating the power output by the accumulator to ensure the accumulator within safe operating limits with regards to temperature. The BMS also controls the path of current flow into and out of the accumulator pack (during charging and discharging cycles).

As per FSAE rule EV.8.3.2 [2], "The AMS [BMS] must have galvanic isolation at every segment-to-segment boundary, as approved in the ESF". The rule dictates the accumulator segment boundaries must be located at galvanic isolation boundaries. Therefore, in the current cell configuration of 104s5p, there are 10 individual groups which must be interfaced with the BMS.

Using the BMS (Orion 108 [19]) from AME22 was originally considered, however it is incapable of servicing all 10 groups as using the Orion 108 with an additional Orion 24 results in up to 9 serviceable groups. Thus, the Orion 108 + 24 setup is still one group short, as demonstrated below in *Figure 62*. To address this, the sub-team elected to source an Orion 2 BMS (Orion 180) [20], which can service up to 16 galvanically isolated groups. Unfortunately, the Orion 180 BMS must still be used to meet rules compliance, despite 6 of the 16 total serviceable groups remaining unused. As a result, the BMS will be larger and heavier than the previous setup featured on AME22. Choosing the larger, heavier BMS was not optimal, but still preferred over developing a fully custom BMS in-house, a task which was deemed much too time-consuming to be achieved during this project's design cycle.

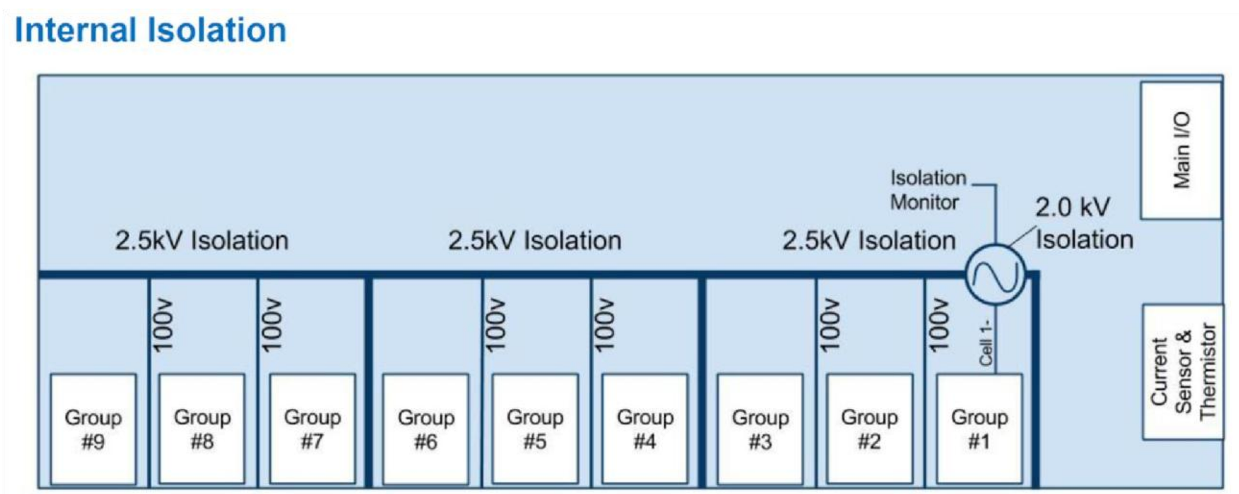


Figure 62: Isolation Diagram for 108 cell system (of which 180 cell diagram may be extrapolated); 108 cell BMS variant only capable of servicing 9 isolated groupings.

7.5.2 Orion BMS 2 (Orion 180)

Figures 63 & 64: Orion BMS 2 (L), BMS kit components (R)

As mentioned previously, usage of this BMS was deemed necessary to design a rules compliant accumulator system. The Orion 180 BMS kit includes all necessary wiring harnesses, current sensors, and CAN adapter to allow computer programming. The BMS also includes a license for Orion's BMS Utility software shown in *Figure 65* which allows for custom profiles to be set, charger integration and control, thermistor readings, and safety control (achieved through charge and discharge limits).

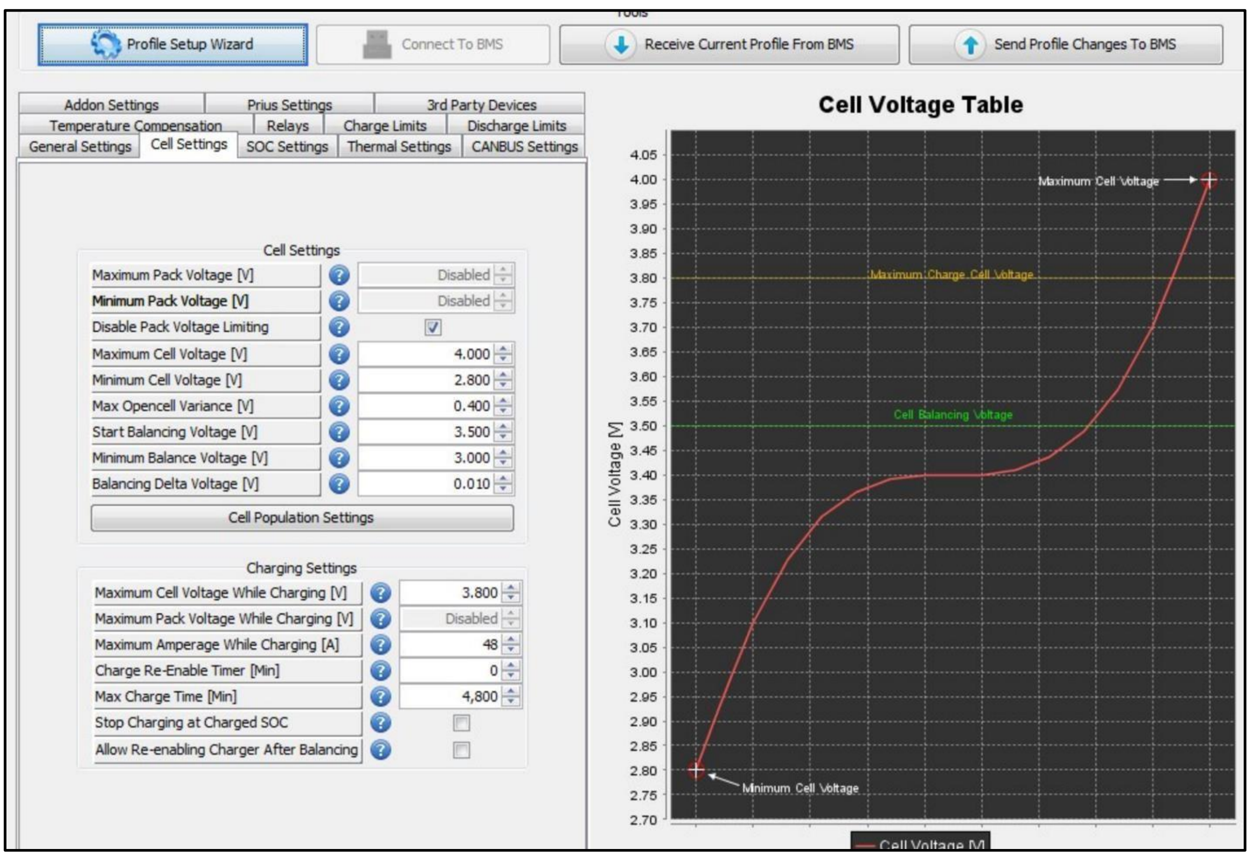


Figure 65: Orion BMS Utility software; Cell Voltage Table plot indicating minimum cell voltage, balancing voltage, maximum charge voltage, and maximum cell voltage

In addition to regulating the cell segments discharge cycles, the BMS must also handle regulation during charge cycles and interfaces directly with the charging unit through the main connector in the BMS wiring harness. The BMS also receives taps pertaining to each cell grouping, as well as the peak voltage. Other interfaces on the BMS include the ground lug which will be connected to the chassis ground, and the thermistor current sensor which is utilized in both charging and discharging cycles. A CAN adapter is utilized to interface with a computer for programming and record viewing and is attached to the main connector port of the BMS. An indepth view of the Orion BMS 2 schematic is provided in *Figure 66*.

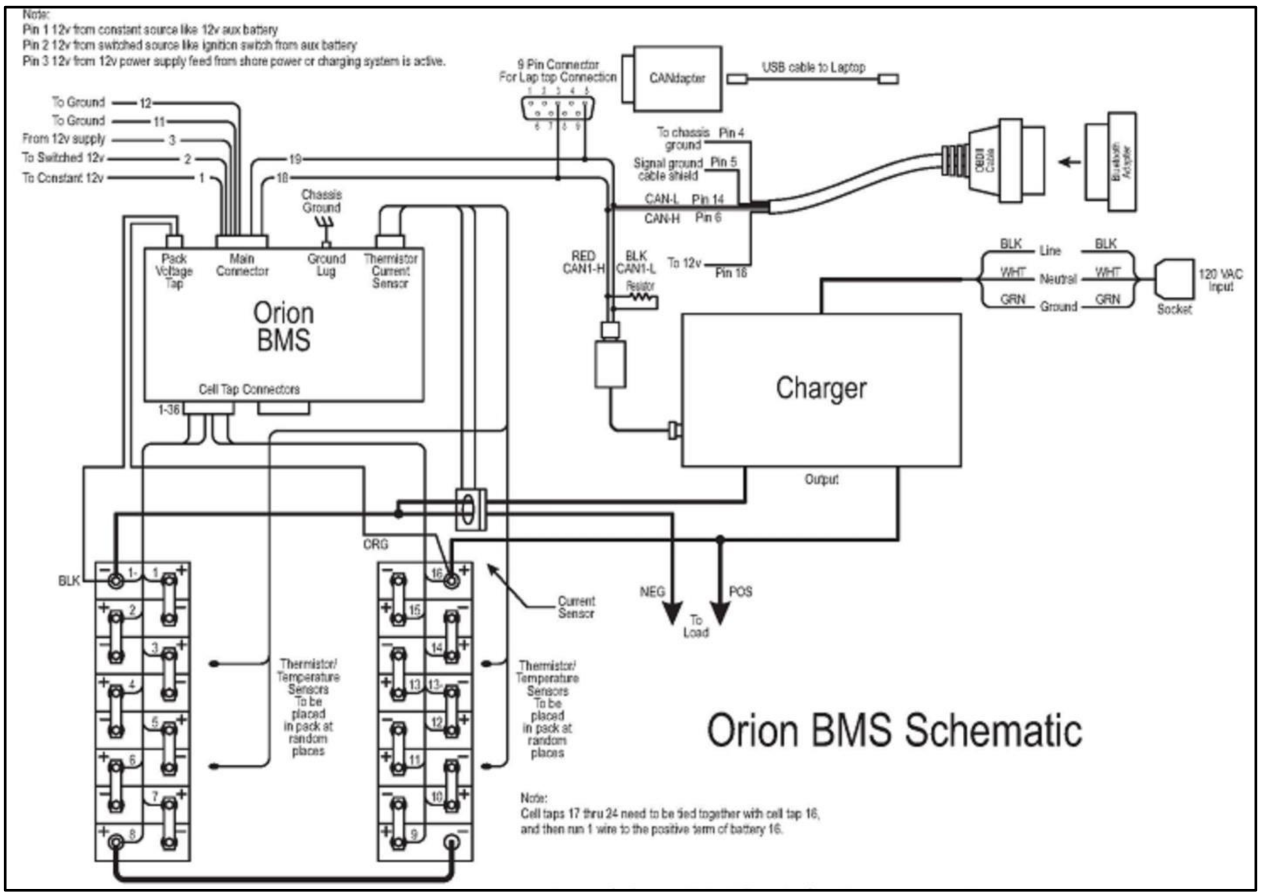


Figure 66: Orion BMS 2 Schematic showing capability of handling up to 16 isolated groupings

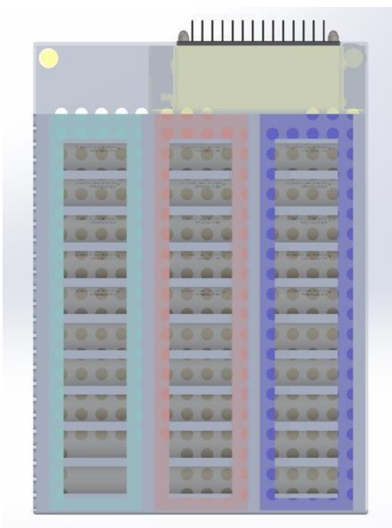
7.6 Accumulator Container

7.6.1 Container Requirements and Function

The main purpose of the accumulator container is to house the accumulator (divided into several segments as defined in sec. 7.5). Additionally, the container must serve as the mounting point for the BMS (see sec. 7.6) and provide access to cell segments for passive air-cooling (see sec. 7.4). To meet the structural requirements defined in the 2023 FSAE rules [2] for the accumulator container, the openings must be less than 25% of each wall. The air streams will be let in through the sides of the container, at which point they will travel across the length of each segment prior to being exhausted through the back of the Y-Long shape by means of an air channel located on the interior of the case. The accumulator container must also be designed such that entry to the container internals can be achieved through the removal or opening of the top lid, which is necessary for segment installation and removal and accumulator maintenance.

7.6.2 Container Design

In accordance with the requirements defined previously and the FSAE 2023 rules [2], the design pictured in *Figure 68* was chosen. To meet packaging requirements for integration with the chassis and minimization of weight, the accumulator container closely traces the shape of the accumulator itself (Y-Long). The design also features a removable lid to allow access to internal components (segments, fuse bars, maintenance plugs, etc.), which is attached to the main body by 4 Metric-Grade 8.8 fasteners (HX-SHCS 0.625 [21])



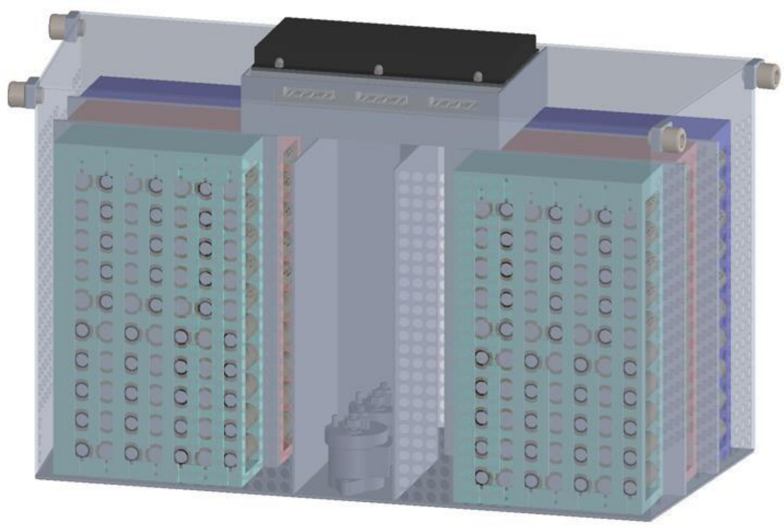


Figure 67 & 68: Top and dimetric views of accumulator container with 6 segments indicated;

[green – 7x10, red – 8x10, and blue – 10x10]

Several materials were considered for the design of the accumulator container including carbon, alloy, and stainless steels, and various aluminum alloys. During the material selection process minimization of predicted mass and maximization of strength (both yield and ultimate tensile) were used as guiding principles. **Table 18** shows the comparisons of the different materials examined.

Table 18: Accumulator container comparison table

Material	Density (g/cc)	Yield Strength (MPa)	Ultimate Strength (MPa	MoE (GPa)	Shear Modulus (GPa)	Elongation at break (%)	Predicted mass (kg)	Predicted cost ($)	LINK
Carbon Steel (AISI 1117)	7.85	305	465	205	80	34	43.175	10.79375	LINK
Alloy Steel (AISI 4130)	7.85	435	670	205	80	25.5	43.175	21.5875	LINK
Stainless Steel (304)	8	215	505	193	77	70	44	118.8	LINK
Aluminum (6061-T6)	2.7	276	310	68.9	26	17	29.7	61.182	LINK
Aluminum (1060-O)	2.705	17	68.9	68.9	26	43	29.755	90.15765	LINK
Aluminum (5093-O)	2.66	145	290	71	26.4	22	29.26	68.1758	LINK

Using **Table 18**, 6061-T6 Aluminum alloy [22] was chosen to be the container material. The container features a thickness of 3.2 [mm] for the top lid, and 2.3 [mm] for the base. To satisfy the rules regarding container structural rigidity, the air ventilation holes feature a diameter of 10 [mm] on all ventilated faces. The final design (*Figure 69*) of the accumulator container has a mass of 57.7 [kg] and a corresponding volume of 0.01152 [m^3].

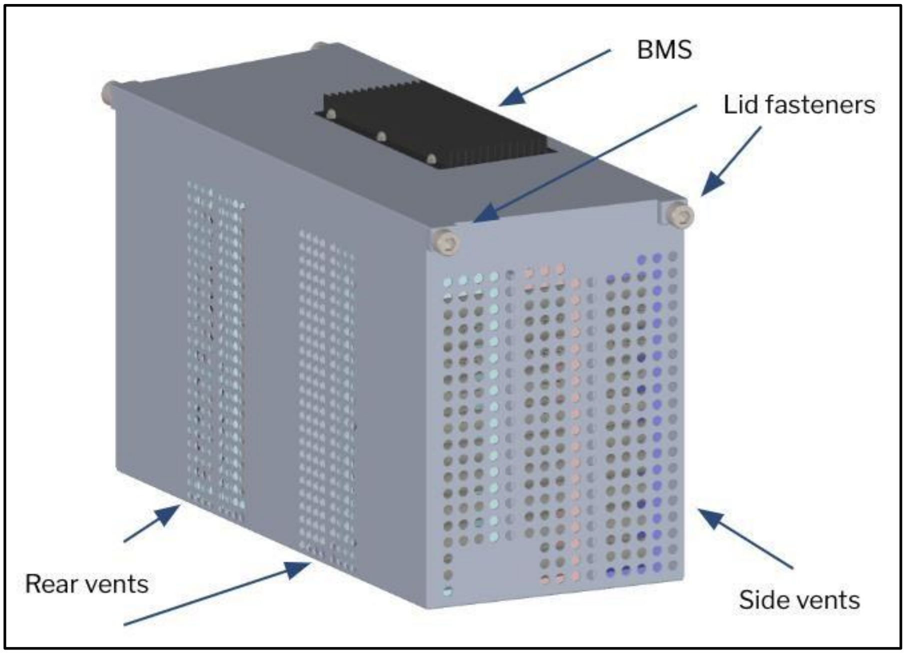
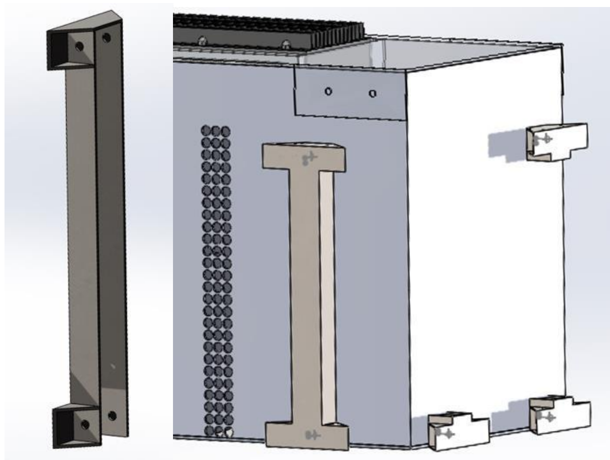


Figure 69: Final accumulator container design with key areas and components indicated

7.6.3 Container Mounting Design

Several mounting methods were considered to effectively mount the accumulator container through either corner-based or mass-based mounting. Corner-based mounting provides increased rigidity for connections located in high stress regions, whilst mass-based mounting allows for simplified bottom entry to the container. The chosen mounting methods consist of brackets, critical fasteners, lock nuts, and tabs (originating from chassis). *Figure 71* includes the current mounting design.



Figures 70 & 71: Container mounting design (L), right half of container showing mounting (R)

9.1.2 Battery Architecture

The largest system developed by the Battery sub-team is the accumulator itself. The Accumulator is made of 6061-T6 Aluminum alloy with an estimated mass of 10.2 [kg] and a corresponding volume of 0.059 [m^3]. The accumulator features a lid mounted BMS, which upon removal will allow for top entry to the accumulator interior in which the accumulator segments are located. The internal walls of the accumulator are lined with Nomex 410 to satisfy fire protection requirements outlined in the 2023 SAE rules [x]. To cool the accumulator, passive air cooling will be utilized with 9.5 [mm] holes comprising 11.5% of the accumulator structure.

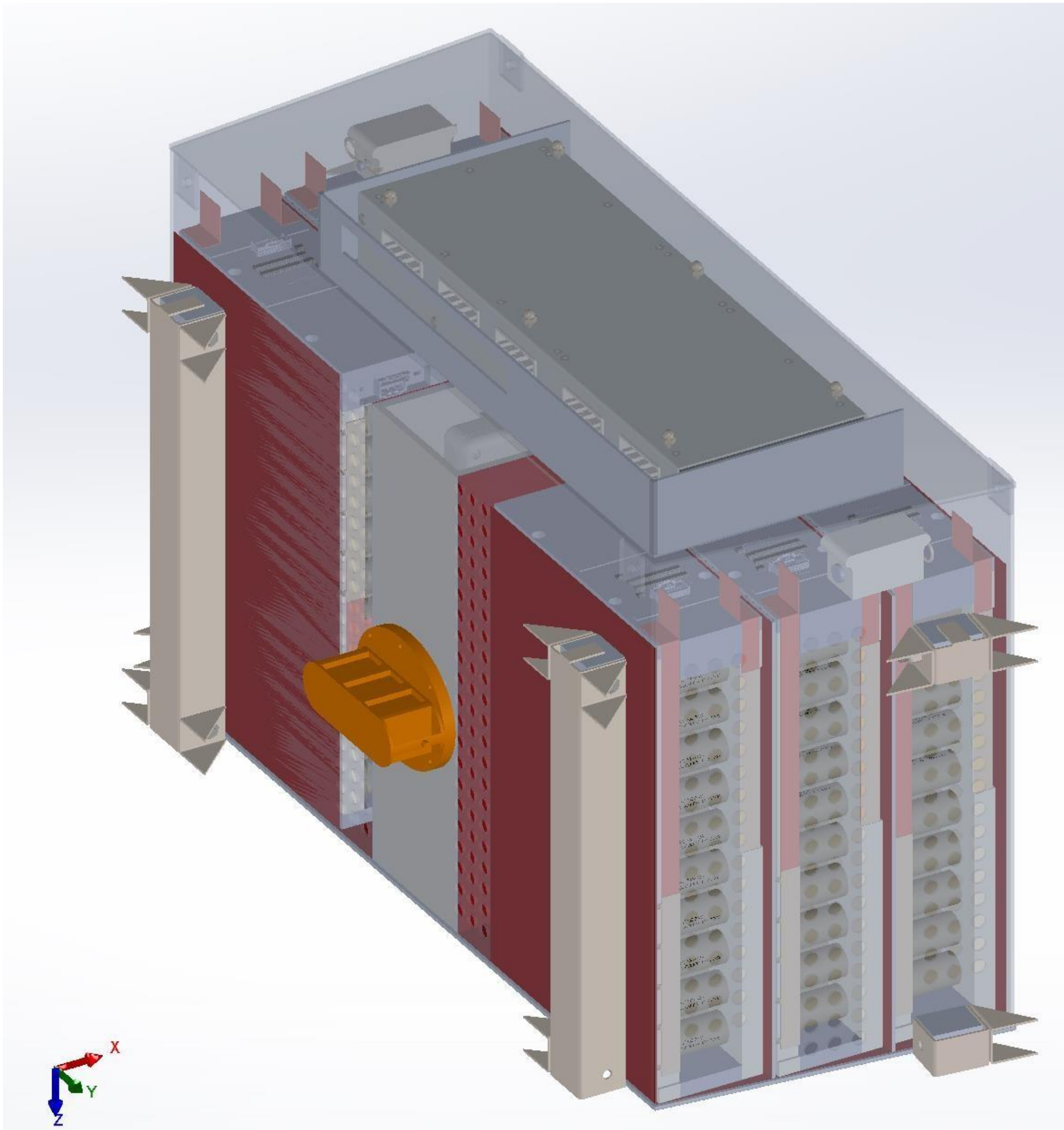


Figure 84: Battery Sub-Team Accumulator Design

9.1.3 Electronics Architecture

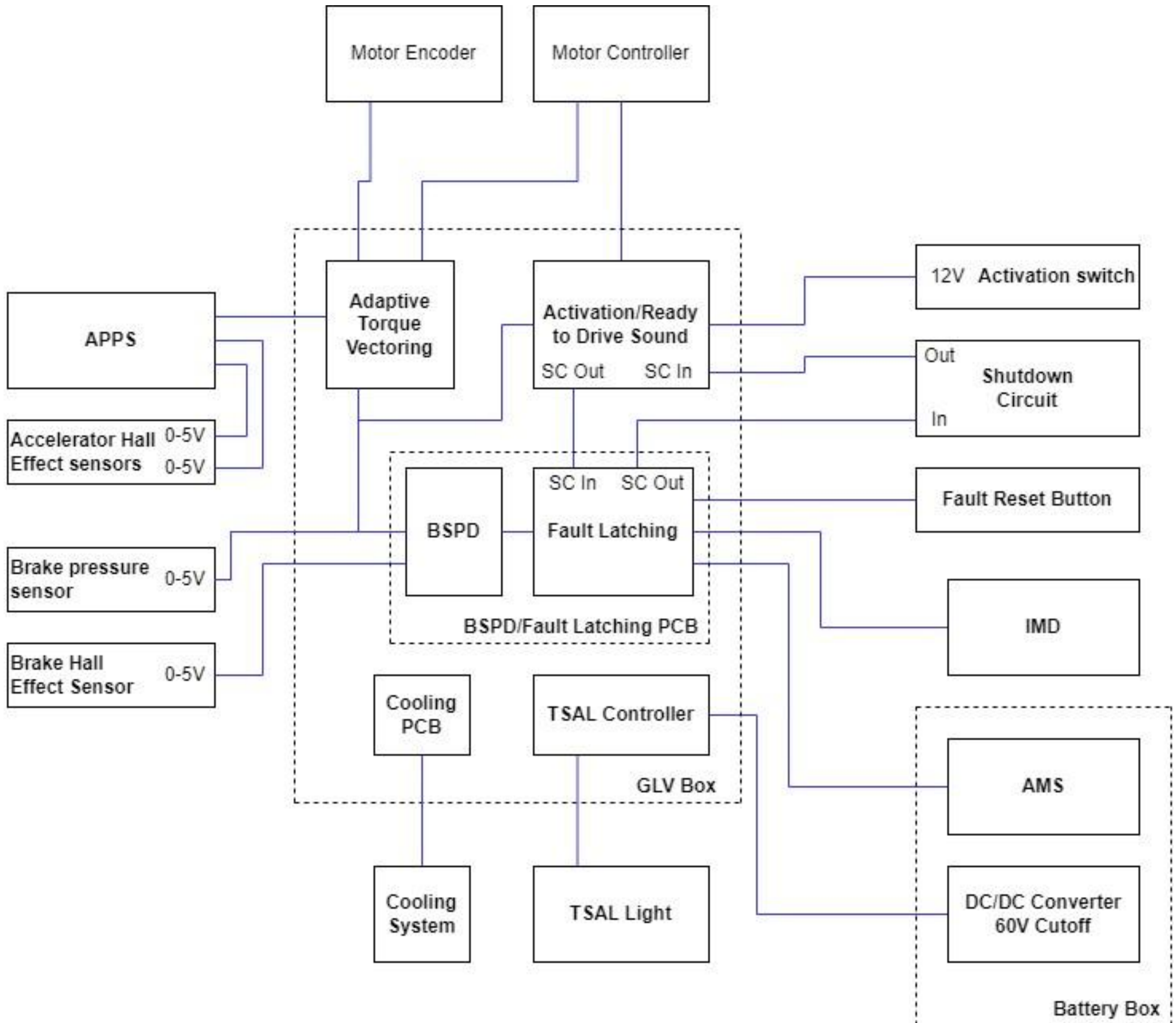


Figure 85: LV System Architecture Diagram

The LV system includes several connections to other sub-teams' components, including sensors and circuit systems. Subsystems such as the APPS and BSPD are involved with Powertrain's drivetrain components through the ATV module which is used to dispatch the Motor Controllers. Additionally, the TSAL is driven by the accumulator system designed by the Battery sub-team.

The Electronics sub-team has its own functionality within the LV system, notably the Shutdown Circuit, DAQ, IMD, AMS, and the ARDS speaker and alert system. Many of the Electronics subsystems are contained within the GLV Box, which itself contains the PCB controlling the BSPD and fault latching capabilities associated with the Shutdown circuit.

9.2.2 Battery Layout

Comprising the accumulator are four 9x10 and two 8x10 segment structures. The number of segments was determined in accordance with maximizing accumulator energy within the predefined trapezoidal shape associated with accumulator placement behind the driver. Air Channels separate the segments and allow for passive air-cooling of the accumulator.

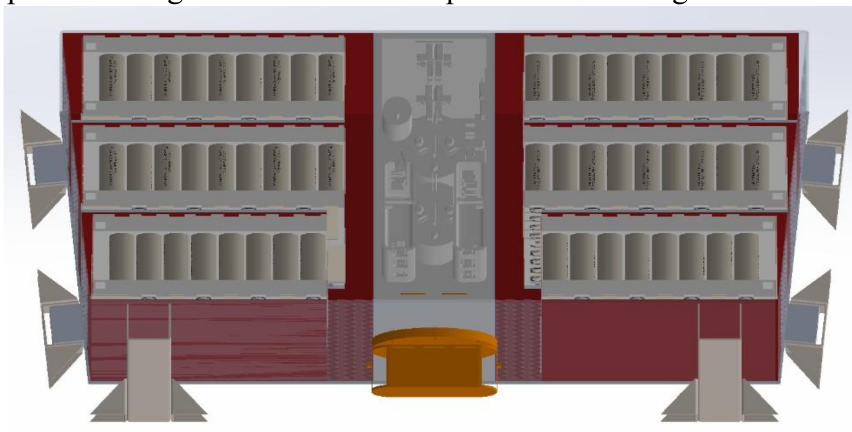


Figure 91: Segments Within Accumulator and Air Channels

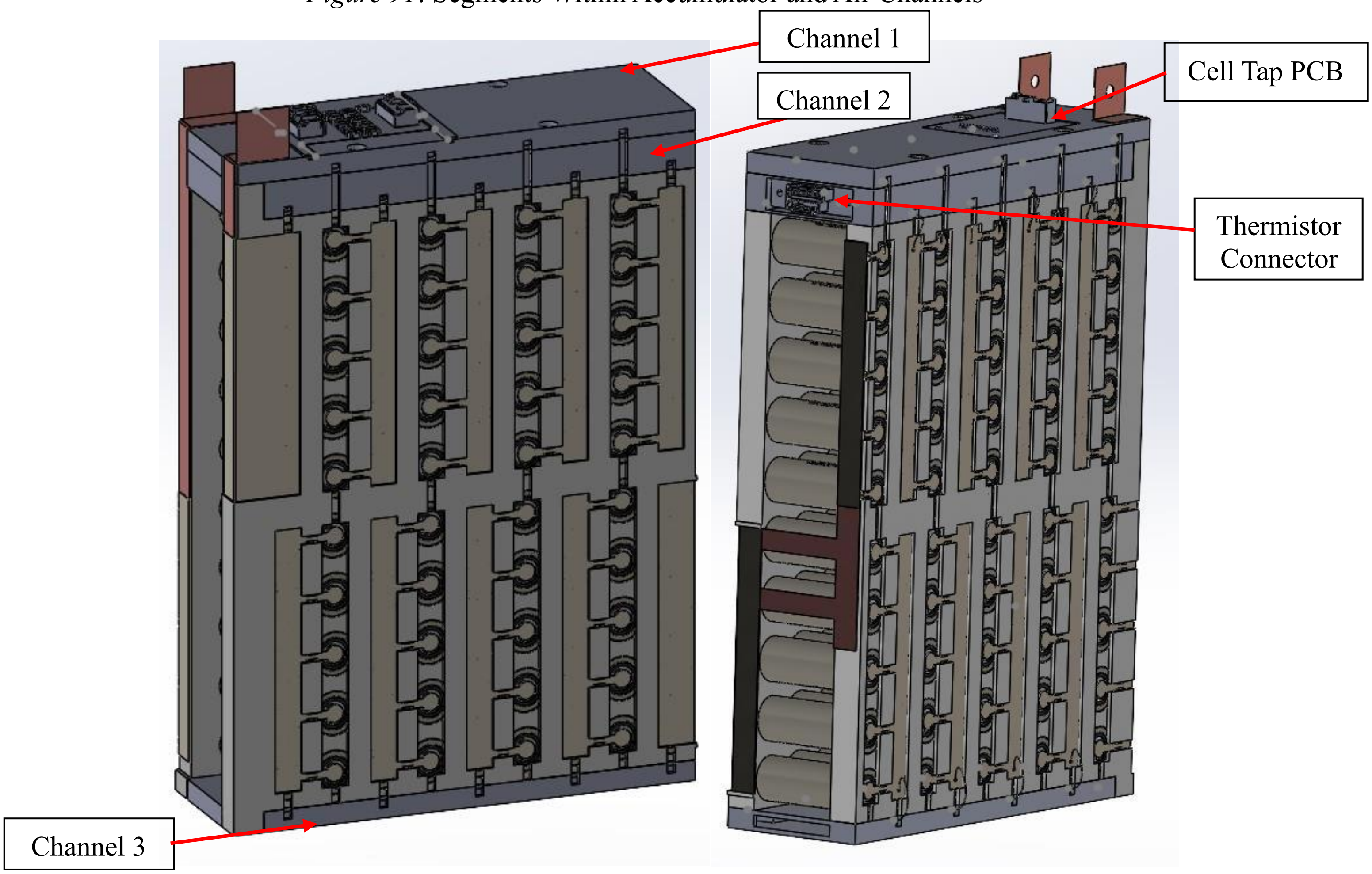


Figure 92: 9x10 Accumulator Segment

The overall design for the segment structure was straightforward since the requirements are driven by what the electrical engineers needed. The segment structures need three channels, one for the cell taps, and two for the thermistor. The channels are listed in figure 92. Channel 1, the cell tap channel, is completely isolated from channels 2 & 3, and have two wires coming down from the top to connect to each series connection. Channel one also had to have a slot for the cell tap PCB which is located on the top and is connected to the BMS. Channels 2&3 are routed along the edge of the segment structure and are connected. Channel 2 was designed to have a slot for the thermistor/BMS connector. The slots along the Segment walls were slotted to accommodate the fuse bars and busbars needed to connect the cells.

PETG and PC-ABS were the two materials that qualified for the segment structure since they both had a deflection temperature rating higher than 60c. The PC-ABS had a much higher rating than PETG but would cost $1300 versus $180 to print with PETG. The main concern was that for testing the temperatures of 60c would be encountered which made PC-ABS a safer option, but the team decided to increase the cooling performance FOS to go with the cheaper PETG option.

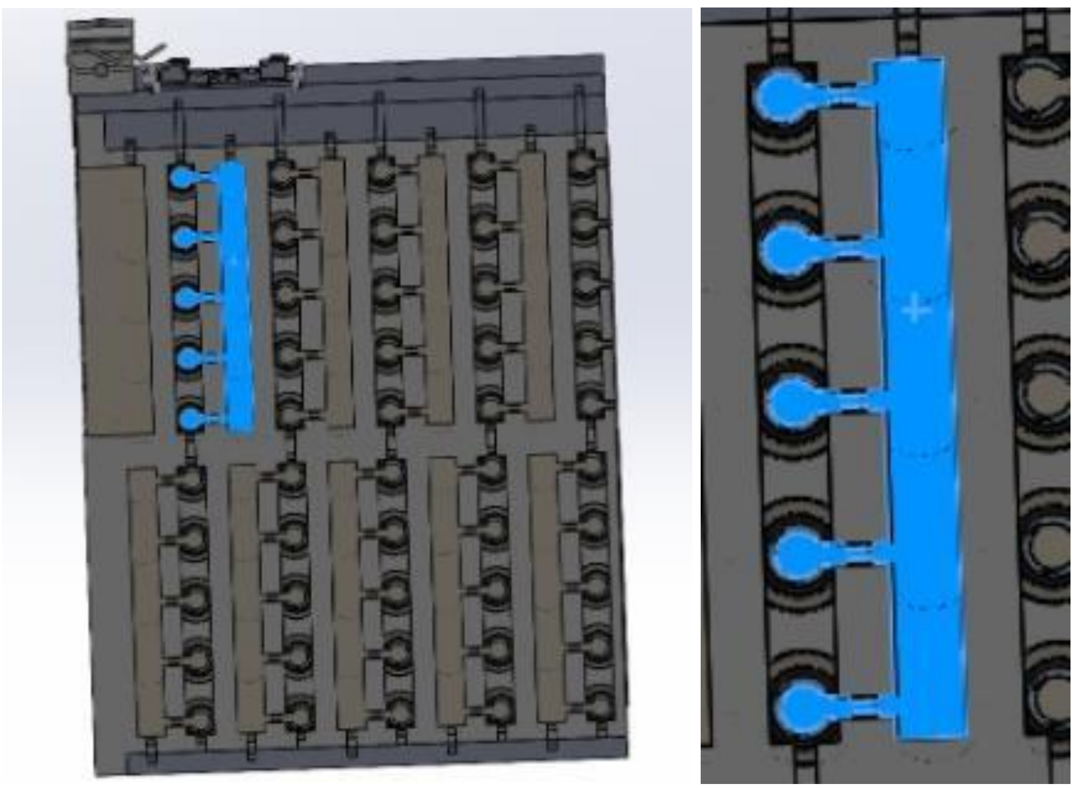


Figure 93: Battery Fusebars shown in blue

Additionally, the Battery accumulator also features the AIRs and Fuses for pack protection. The
AIRs used are LEV200A4NA, which feature electrical capability of up to 500 [A] and 900 [VDC] along with a coil voltage of 12 [V] and a low coil current of 1.1 [A]. The selected fuses are L50QS100.T which handle up to 100 [A], 500 [VDC] with a 50 [kA] breaking capacity.

Figure 94 & 95: Battery AIR (L) and Fuse (R) Selections for Accumulator Pack Protections

9.3.2 Battery Technical Analysis

The battery pack/accumulator mounts were built with the team philosophy in mind since one of the teams' main goals was to maximize testing time. The slot mechanism acts as blind mate that aligns the 10 accumulator attachment points when the pack is in the correct orientation.

This design saves significant time when removing and installing the accumulator from the chassis versus manually aligning the mounts which is traditionally done by most teams.

The There are two types of steels the team considered for our mounts. One being ASTM 513 and the other being 4130 steels. The mounts must be made from carbon steel since we plan to weld the mounts to the chassis, which is made from 4130 steel. The accepted maximum yield strength is 220 MPa ad 517 MPa for 513 and 4130 respectively. Since 513 had a lower yield strength, more material needed to be used to match the performance of the 4130 steels. The decision to go with 4130 steels for our mounts meant a lower weight and less volumetric space used so our decision for 4130 steel was justified.

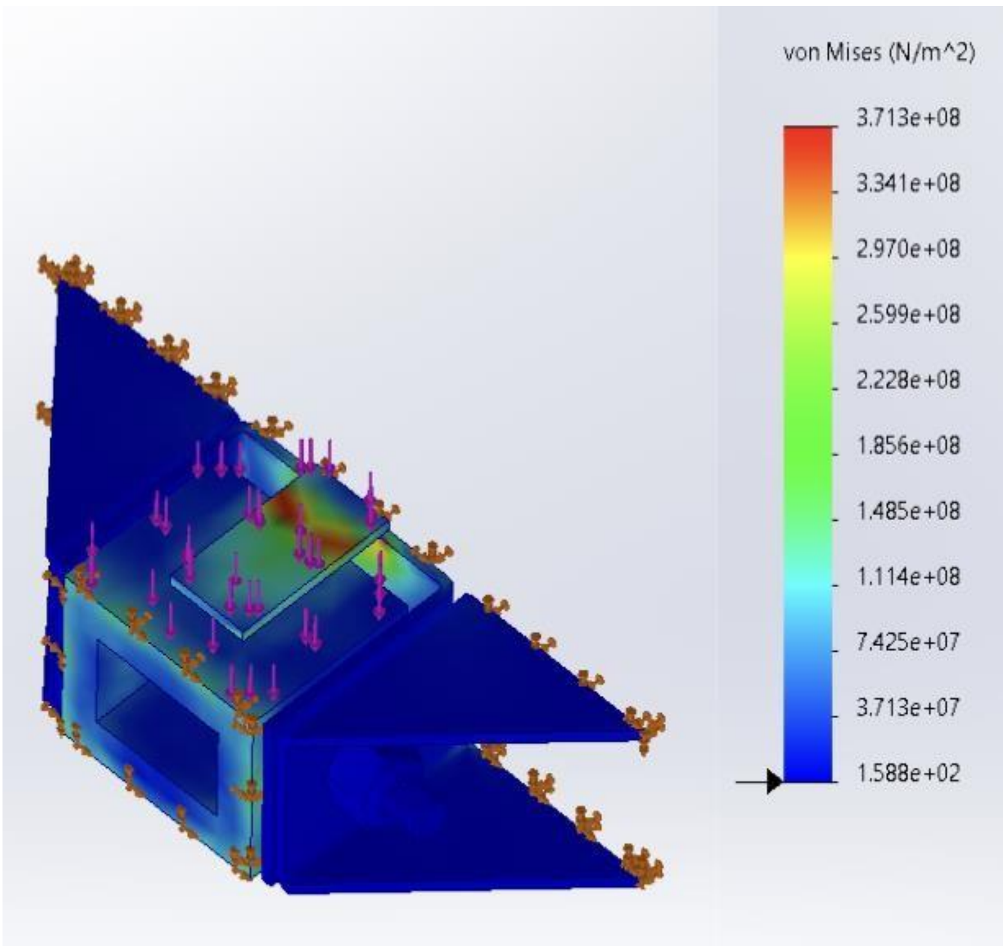


Figure 106: Accumulator Container FEA

The accumulator tabs are made from the same aluminum as the accumulator container, 6061-T6. There are a variety of tabs the team could choose from such as U-cross section, H-cross section, and Square cross section. The team ultimately went with a square cross section as this was the lightest option without the need for a gusset. The U-cross section and the U-cross section allow for better weight savings but the need for a gusset meant the U-channel alignment mechanism from the chassis side mounts would cause an obstruction.

SolidWorks FEA was used to determine the integrity of our mounting designs. SolidWorks has preset yield strengths for the 4130 steel and 6061-T6 materials we planned to use but we decided to input the rated data from the manufacturing website to give us a valid result. The bolts were given a pre-torque value of 16.9 N-m, and that was found from the pre-torque calculator the team made. The loading conditions are purely based on rules since the driving requirement is that any accumulator connection point must endure 15kn of force. The mounts were to endure theses forces in the X, Y, and Z directions of the mounts. The welds for the tabs and the mounts were modeled by having all the coincident edge as fixed to simulate the weld. Our results gave us an FOS of 1.67 and the highest stress concentrations occur at the weld point as shown in the isometric view in *Figure 107*. This figure depicts the bottom mount undergoing a 15kn load in the vertical direction since this was the highest stress mount. Overall, the team would like to keep our FOS between 1.1-1.2 since we are trying not to over design our elements, and to keep our components as light as possible while maintain the necessary performance.

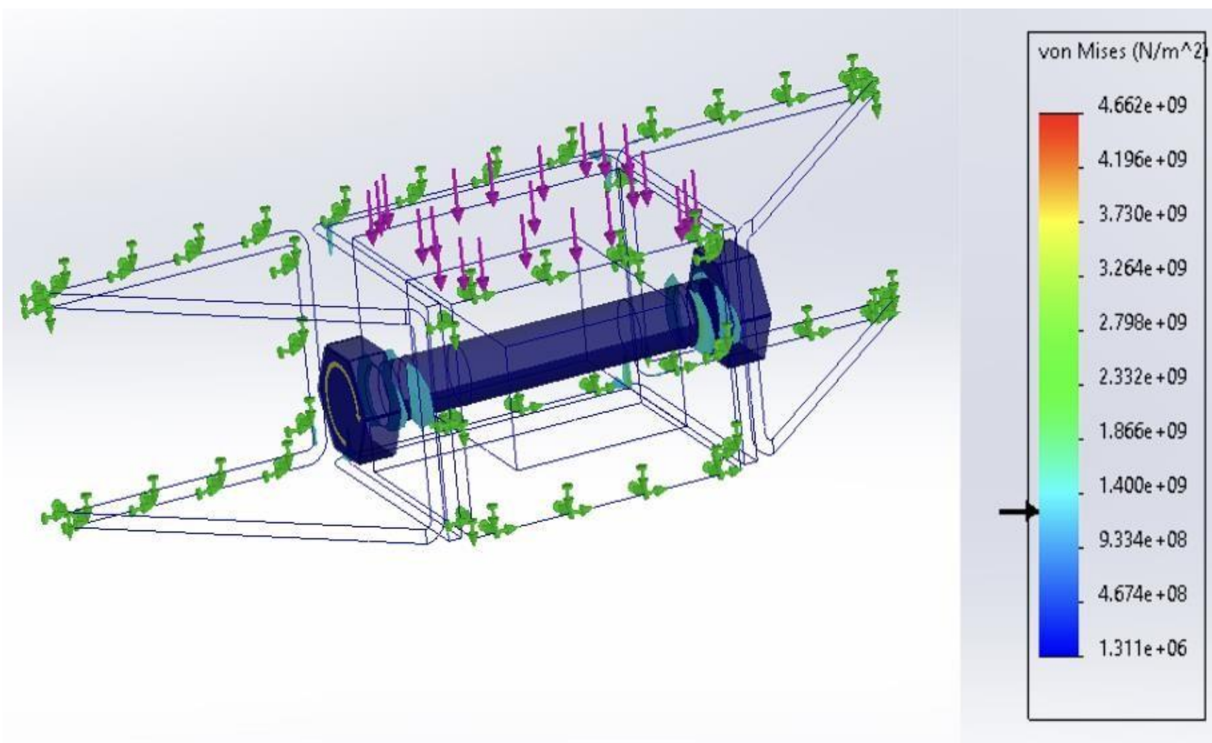


Figure 107: Accumulator Mounting FEA

9.4 Preliminary Product Risk Analysis

9.4.1 Powertrain & Battery Subsystems FMEA

Table 20: Powertrain & Battery Subsystems FMEA

Sub-team & Component	Potential Failure Mode	Potential Effects of Failure	Severity (S)	Potential Causes and Mechanisms of Failure	Occurrence (O)	Current Design Controls Test	Detection (D)	Recommended Action	RPN
Battery, Case	Warping, cracking / breaking	Loss of accumulator	10	Fatigue or overloading	2	FEA, destructive testing if budget permits	3	Use higher than normal FOS	**60**
Battery, Mounts	Warping, cracking / breaking	Accumulator becomes loose in chassis	9	Fatigue or overloading	1	FEA, destructive testing if budget permits	2	Use higher than normal FOS	**18**
Battery, Segments	Melting	Short circuit	7	Inadequate cooling	4	Small scale pre-testing	2	Use temperature monitoring system	**56**
Battery, cell fuses	Fails to trip	Cells overheat and start fire	9	Improperly designed fuse, manufacturing defects	3	Thorough pretesting of fuses	2	Incorporate thermal shutdown system using thermocouples.	**54**
Powertrain, Hub	Cracking and shearing	Catastrophic failure of suspension	9	Fatigue or overloading	4	FEA, Destructive testing if budget permits	2	Use higher than normal FOS	**72**
Powertrain, Casing	Cracking and breaking	Catastrophic failure of suspension	9	Fatigue or overloading	2	FEA, Destructive testing if budget permits	4	Use higher than normal FOS	**72**
Powertrain, Carrier	Cracking and breaking	Lack of torque transmission, failure of gearset	9	Fatigue or overloading	4	FEA, Destructive testing if budget permits	2	Use higher than normal FOS	**72**
Powertrain, Cooling System	Leak in cooling lines	Overheating of motor and controller	6	Incorrectly attached tubing	4	Extra checks to make sure all components are compatible	4	Utilize temperature and cooling sensors within cooling system	**96**
Powertrain, Motors	Motor becomes eccentric	excess loads on wheels	4	Manufacturing defect, improper loading	5	Preliminary testing	2	Benchmark the motor initially and record data for comparison	**40**
Powertrain, Torque Vectoring System	Failure of boards/wiring	Motor does not spin	3	mistake on board, wear, and tear	6	Preliminary testing preinstallation	3	Conduct thorough testing of all modes	**54**

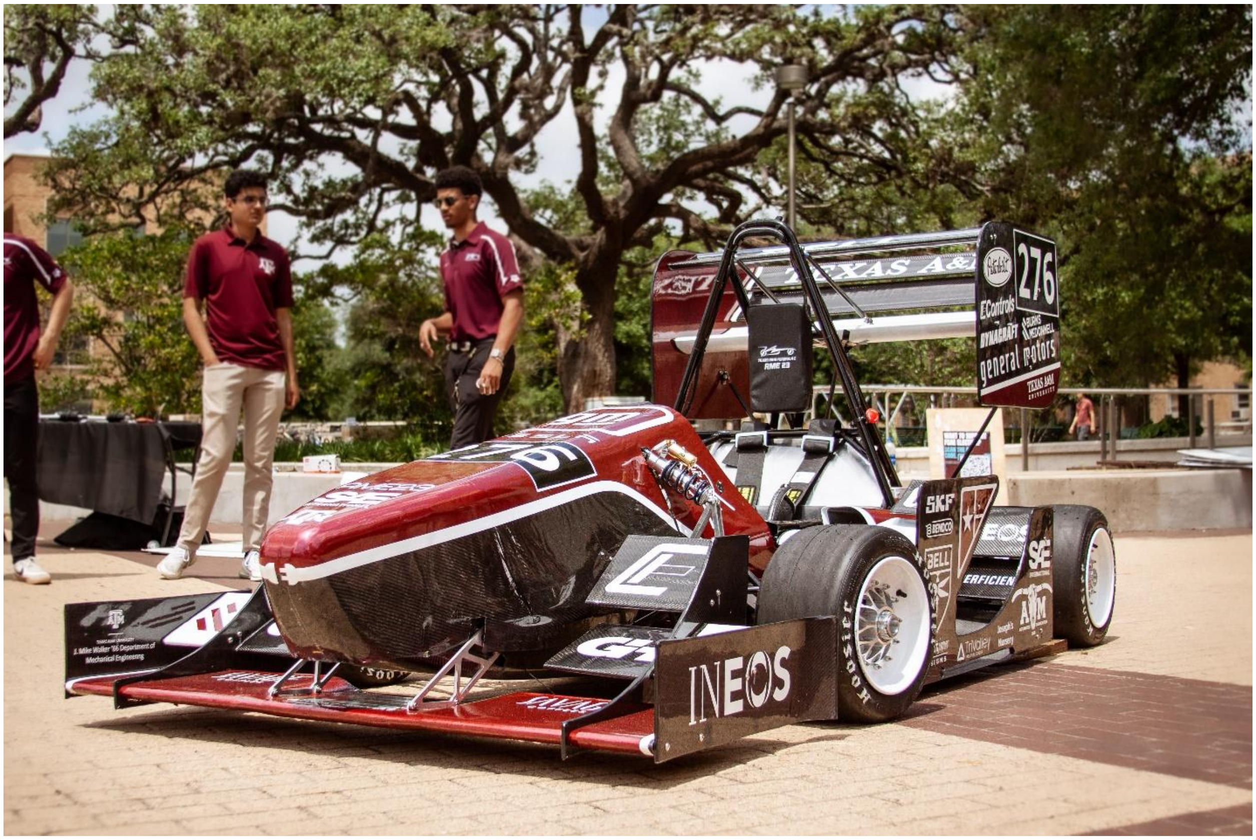
11.2.2 Battery BOM

Item	Unit Cost	Quantity	Total Cost	Expected Leadtime	Off Shelf/ Custom	Need by:
Cells	0	600	0	6-8w	OTS	Dec

Container Material	370	1	370	2w	OTS	Dec
BMS	0	1	0	0d	OTS	Nov
Nomex	16.08	3	48.24	5d	OTS	Jan
GLV Wire	15.99	1	15.99	1w	OTS	Dec
Bus Bar	48.77	10	487.7	5d	OTS	Dec
TS connector	48.77	2	97.54		OTS	Jan
Tap & Thermistor Connectors	204.95	1	204.95	2w	OTS	Dec
PETG	18.99	6	113.94	5d	OTS	Nov
AIRs	139.27	2	278.54	2w	OTS	Dec
Fuse	105.4	1	105.4	1w	OTS	Jan
HVIL Components	125.33	1	125.33	4w	Custom	Dec
IMD	25	1	25	1w	OTS	Jan
TS Wire	12.99	50	649.5	2w	OTS	Jan
Fasteners	36	1	36	5d	OTS	Dec
BMS Harnesses	58	1	58	1w	OTS	Nov
Thermistors	65	5	325	2w	OTS	Nov
Thermal Expansion Module	216	1	216	1w	OTS	Nov
SOC Display	75	1	75	2w	OTS	Nov
BMS CAN Adapter	75	1	75	2w	OTS	Nov
TOTAL			**3307.13**			

Figure 112: Battery BOM

Figure 116: *Solidworks* Renders of Current Full Vehicle Design with Potential Livery Designs



276
general motors
SKF
BELL
INEOS


SKF
HIGH VOLTAGE
Hoosier
TriValley
Joseph's
Nursery
AUTOMATED
FABRICATION
LAB

https://tamuformulaelectric.com/ame23